CHAPTER 03

# Bonding

## Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 6adaab58d2ab9952457d9fe2d9a446aa1c3660cd16105c2bcc3f278bc11f538e

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

### Coverage checklist

ch03:exercise:quantum | quantum:stage-1, quantum:stage-2, quantum:stage-3

ch03:exercise:bond-count | bond-count:stage-1, bond-count:stage-2, bond-count:stage-3

ch03:exercise:amide-resonance | amide-resonance:stage-1, amide-resonance:stage-2, amide-resonance:stage-3

ch03:lesson:orbitals | orbitals:rule, orbitals:cue, orbitals:trap, orbitals:app, orbitals:step-1, orbitals:step-2

ch03:lesson:sigma-pi | sigma-pi:rule, sigma-pi:cue, sigma-pi:trap, sigma-pi:app, sigma-pi:step-1, sigma-pi:step-2

ch03:lesson:hybridization | hybridization:rule, hybridization:cue, hybridization:trap, hybridization:app, hybridization:step-1, hybridization:step-2, hybridization:step-3

ch03:molecule:ethene | ethene

ch03:molecule:ethyne | ethyne

ch03:molecule:acetamide | acetamide

ch03:molecule:acetaldehyde | acetaldehyde

ch03:molecule:acetate | acetate

ch03:curriculum | scope-1, scope-2, scope-3

# Curriculum scope

scope-1: 3.1 Atomic Orbitals and Quantum Numbers

scope-2: 3.2 Molecular Orbitals

scope-3: 3.3 Hybridization

Objective: Apply allowed quantum numbers

Objective: Count sigma and pi bonds

Objective: Explain hybridization and resonance

Supplied Kaplan curriculum: Chapter 3; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter03b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

# orbitals | Atomic orbitals and quantum numbers

Orbitals describe probability, not paths

## orbitals:rule | Core Rules

n = 1,2,… sets the shell. l = 0,…,n−1 sets the subshell (s,p,d,f). ml = −l,…,+l labels orbitals; ms = ±1/2 labels electron spin. Each orbital holds at most two electrons with opposite spins.

## orbitals:cue | Memory Cues

One s orbital; three p orbitals; five d orbitals. A p subshell can hold six electrons.

## orbitals:trap | Common Mistakes / MCAT Traps

Orbital phase signs are wavefunction signs, not positive and negative electric charges. Electrons do not orbit like planets.

## orbitals:app | MCAT Applications

Carbon bonding uses valence orbitals from the second shell. Hybridization is a model for bonding and geometry, not a change in the allowed quantum numbers.

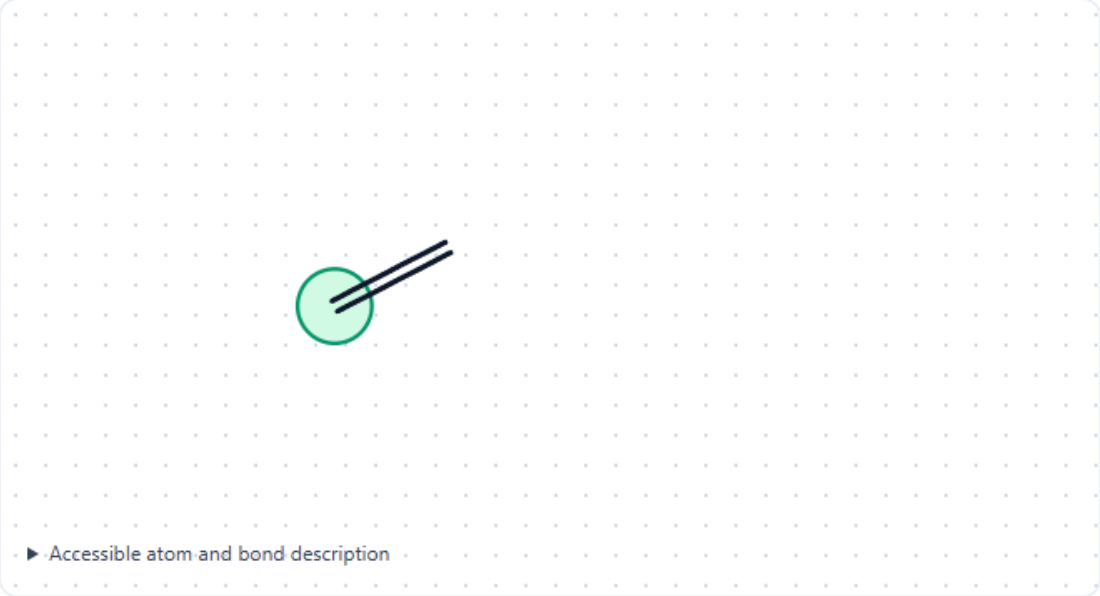
## orbitals:step-1 | 1. Count valence electrons

An isolated neutral carbon atom has configuration $1s^2 2s^2 2p^2$. Its four valence electrons occupy the n=2 shell. In molecules, orbitals combine and mix to describe bonding.

Displayed molecule ID: ethene

{"highlights": {"prin": [0]}, "revealName": false}

Screenshot: orbitals-step-1



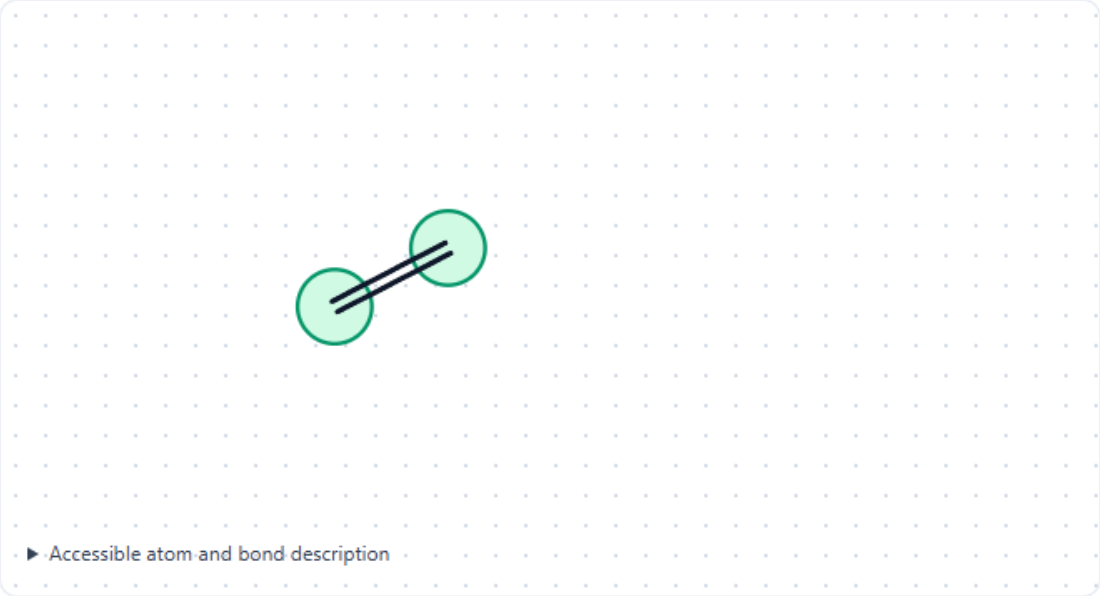

## orbitals:step-2 | 2. Fill degenerate orbitals

Hund’s rule places electrons singly with parallel spins in degenerate orbitals before pairing. Pauli exclusion forbids two electrons in an atom from sharing all four quantum numbers.

Displayed molecule ID: ethene

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Screenshot: orbitals-step-2



# sigma-pi | Sigma, pi, and molecular orbitals

End-on vs. side-on overlap

## sigma-pi:rule | Core Rules

A sigma bond has electron density along the internuclear axis. A pi bond comes from side-on p-orbital overlap. A double bond has one sigma and one pi; a triple bond has one sigma and two pi. Constructive overlap forms a lower-energy bonding MO; destructive overlap gives an antibonding MO with an added node.

## sigma-pi:cue | Memory Cues

Bond order = (bonding electrons − antibonding electrons)/2 in the simple MO model. More bond order usually means shorter, stronger bonds for a given atom pair.

## sigma-pi:trap | Common Mistakes / MCAT Traps

A pi bond restricts rotation because twisting disrupts overlap. Bond rotation is possible only by paying an energy cost; do not imagine a double bond freely rotating at room temperature.

## sigma-pi:app | MCAT Applications

Carbonyl addition converts a pi bond into sigma bonds. Following sigma and pi electrons helps explain the change from planar to tetrahedral geometry.

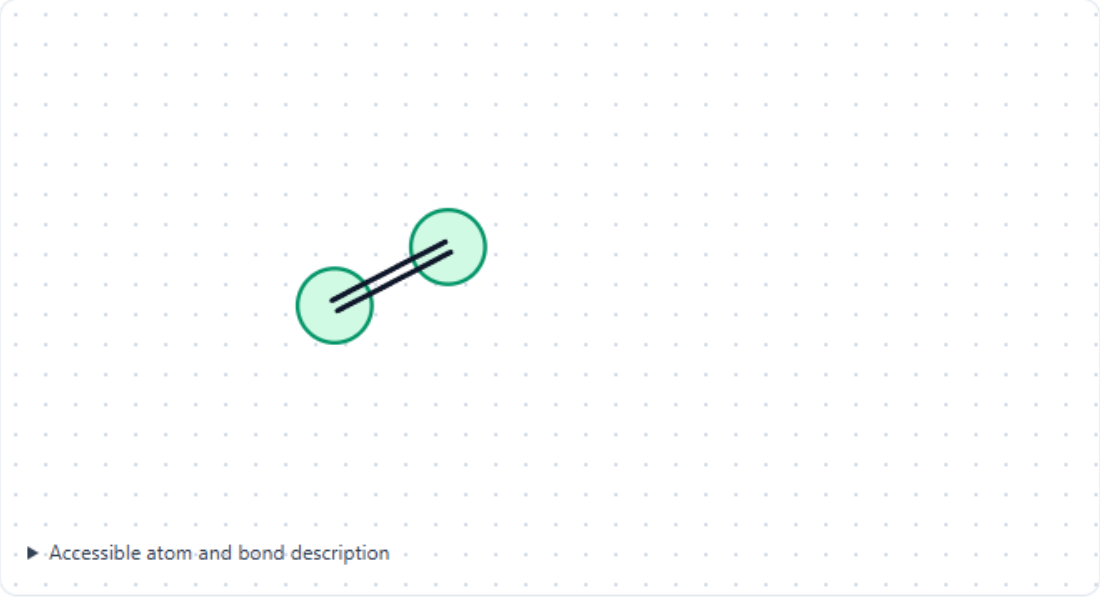
## sigma-pi:step-1 | 1. Double bond inventory

Ethene has one C–C sigma bond, one C–C pi bond, and four C–H sigma bonds: five sigma and one pi overall.

Displayed molecule ID: ethene

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Screenshot: sigma-pi-step-1



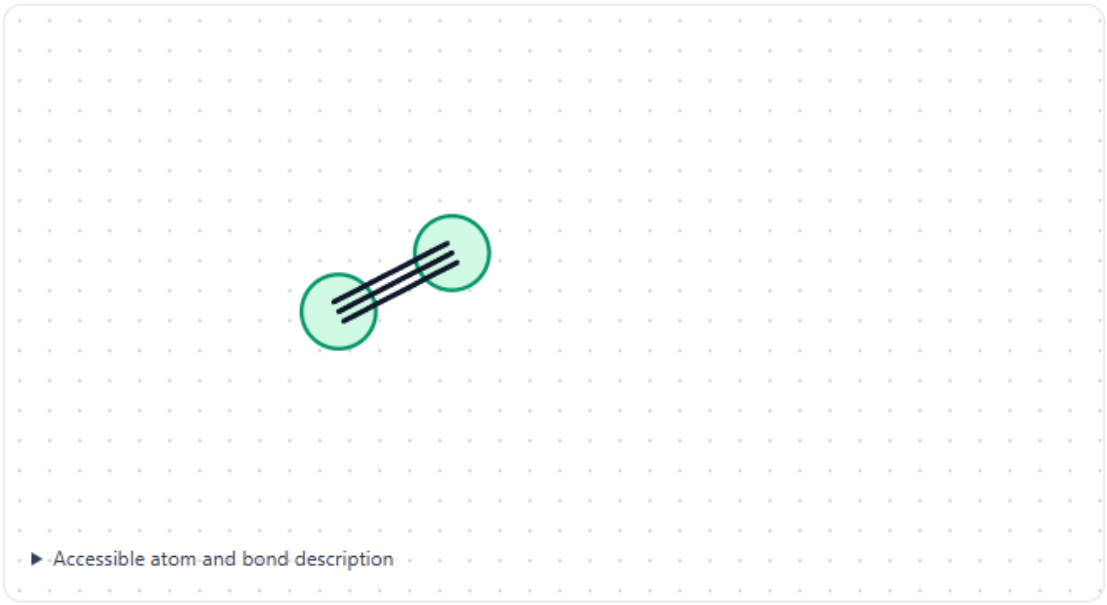

## sigma-pi:step-2 | 2. Triple bond inventory

Ethyne has one C–C sigma plus two perpendicular pi bonds. Add two C–H sigma bonds: three sigma and two pi overall.

Displayed molecule ID: ethyne

{"highlights": {"prin": [0, 1]}, "revealName": true}

Screenshot: sigma-pi-step-2



# hybridization | Hybridization, geometry, and resonance

Count domains, then check conjugation

## hybridization:rule | Core Rules

Typical tetrahedral carbon is $sp^3$ (about 109.5°), trigonal planar carbon is $sp^2$ (about 120°), and linear carbon is sp (180°). The s character is 25%, 33%, and 50%, respectively. Resonance contributors differ only in electron placement, not atom connectivity.

## hybridization:cue | Memory Cues

For ordinary carbon: four electron domains → $sp^3$; three → $sp^2$; two → sp. A multiple bond counts as one domain.

## hybridization:trap | Common Mistakes / MCAT Traps

Do not mechanically label an amide nitrogen $sp^3$ just because it has three sigma bonds and a lone pair. Its lone pair occupies a p orbital conjugated with the carbonyl, making the amide approximately planar.

## hybridization:app | MCAT Applications

Peptide-bond resonance restricts C–N rotation. Delocalized charge explains carboxylate stability and why its two C–O bonds are equivalent in the resonance hybrid.

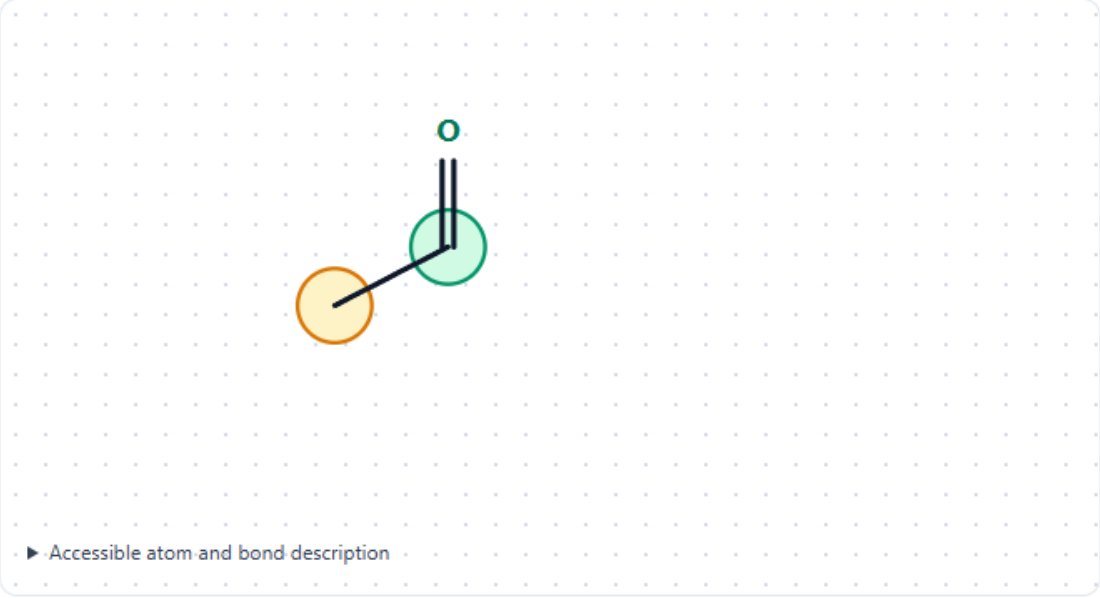
## hybridization:step-1 | 1. Carbonyl geometry

The carbonyl carbon has three sigma-bond directions and a p orbital for the pi bond: $sp^2$ and approximately planar. Its methyl neighbor is $sp^3$.

Displayed molecule ID: acetaldehyde

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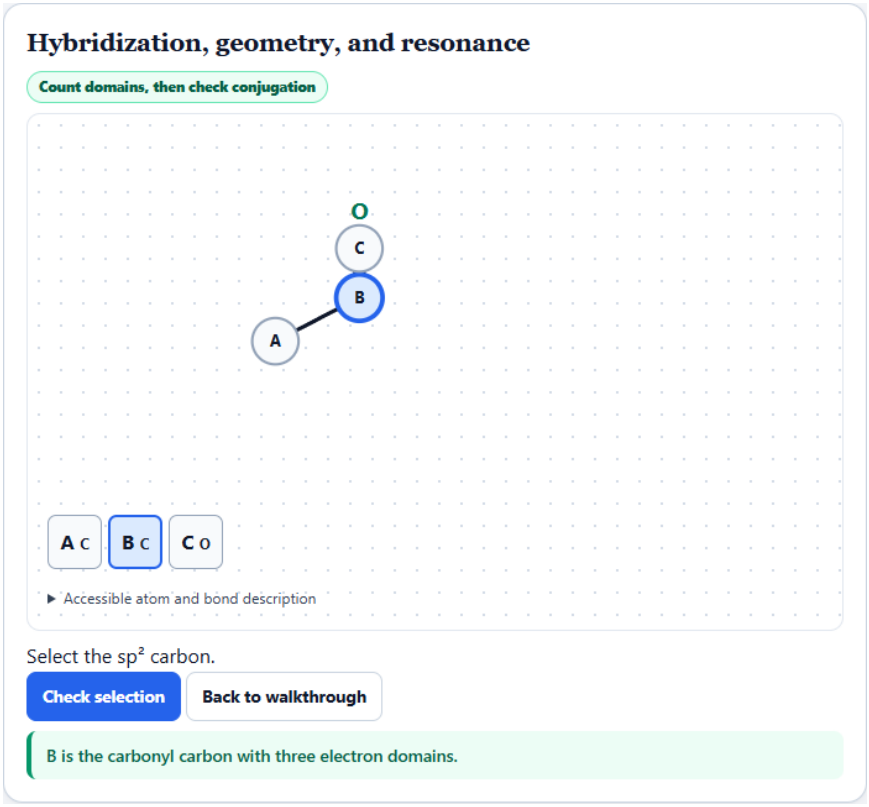
Screenshot: hybridization-step-1




Embedded challenge: Select the $sp^2$ carbon.

{"prompt": "Select the sp² carbon.", "answer": [1], "explanation": "B is the carbonyl carbon with three electron domains."}

Screenshot: hybridization-step-1-challenge



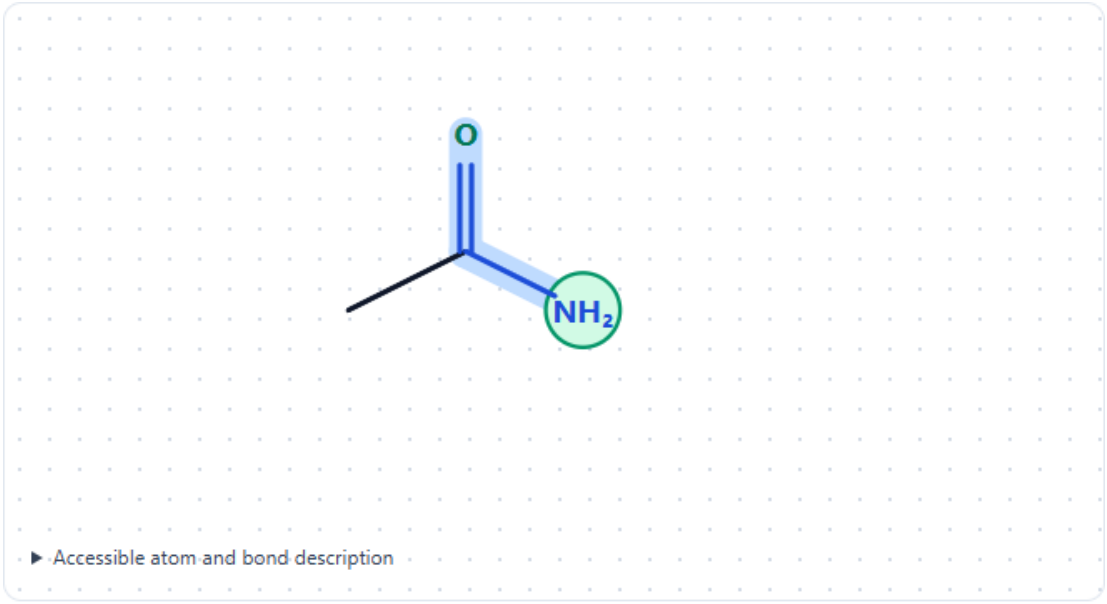

## hybridization:step-2 | 2. Amide conjugation

Nitrogen donates lone-pair electron density into the carbonyl system. The C–N bond has partial double-bond character; amide nitrogen is much less basic than a typical amine.

Displayed molecule ID: acetamide

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Screenshot: hybridization-step-2



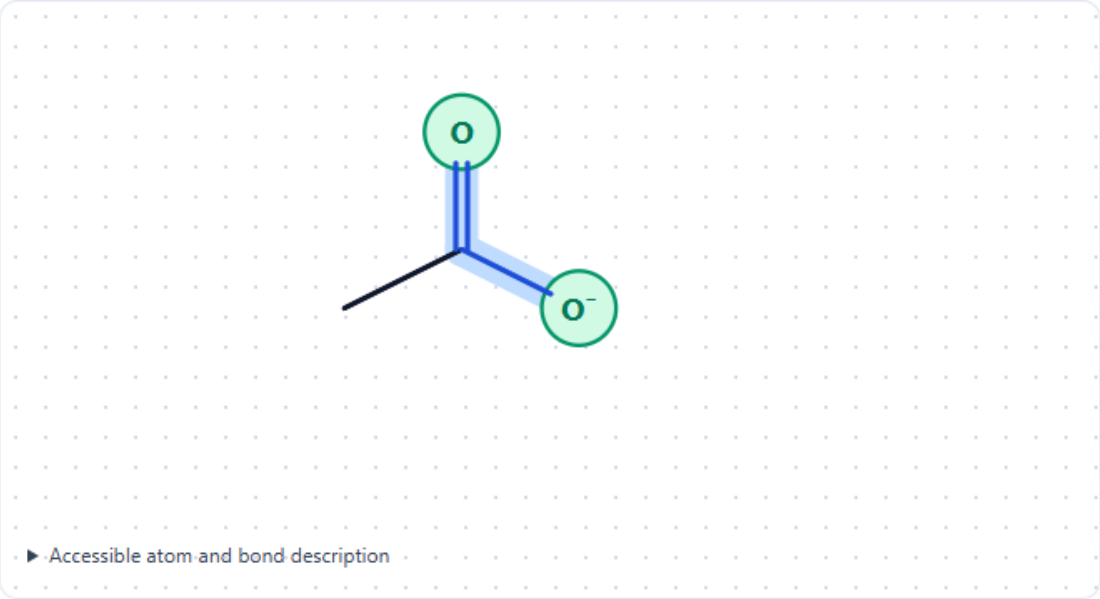
## hybridization:step-3 | 3. Resonance hybrid

This drawing is one contributor of acetate. The other exchanges the two oxygen bond orders and the negative charge. The actual ion has delocalized charge; it does not flip back and forth between drawings.

Displayed molecule ID: acetate

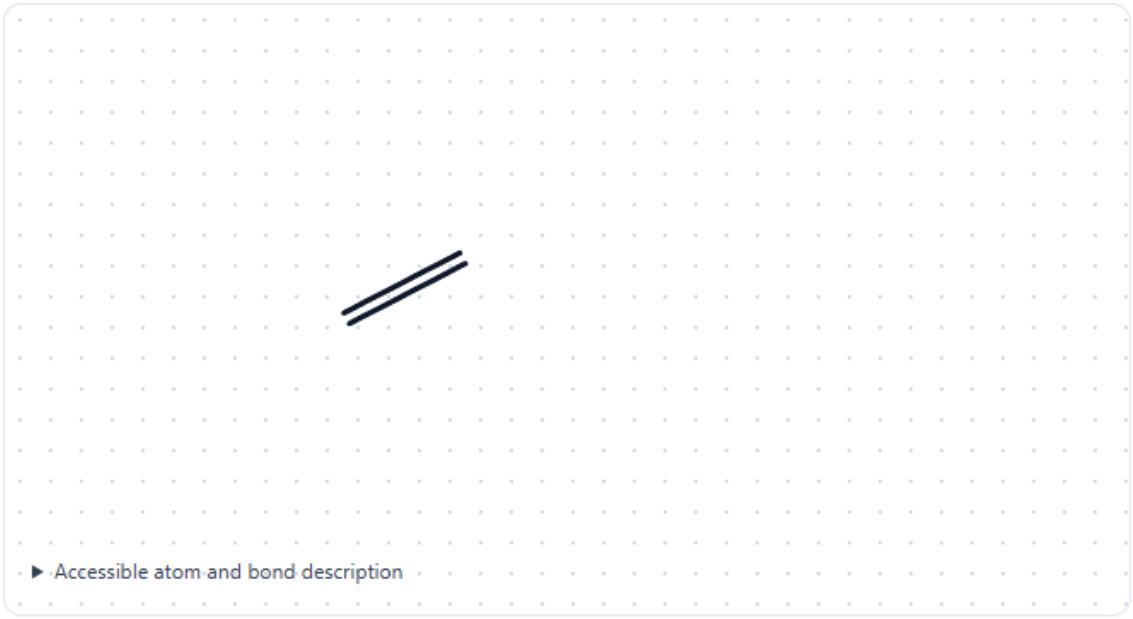
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Screenshot: hybridization-step-3



# quantum | Check allowed quantum numbers

Screenshot: quantum-stage-1-molecule




## quantum:stage-1 | For a 2p electron, which n and l pair is allowed?

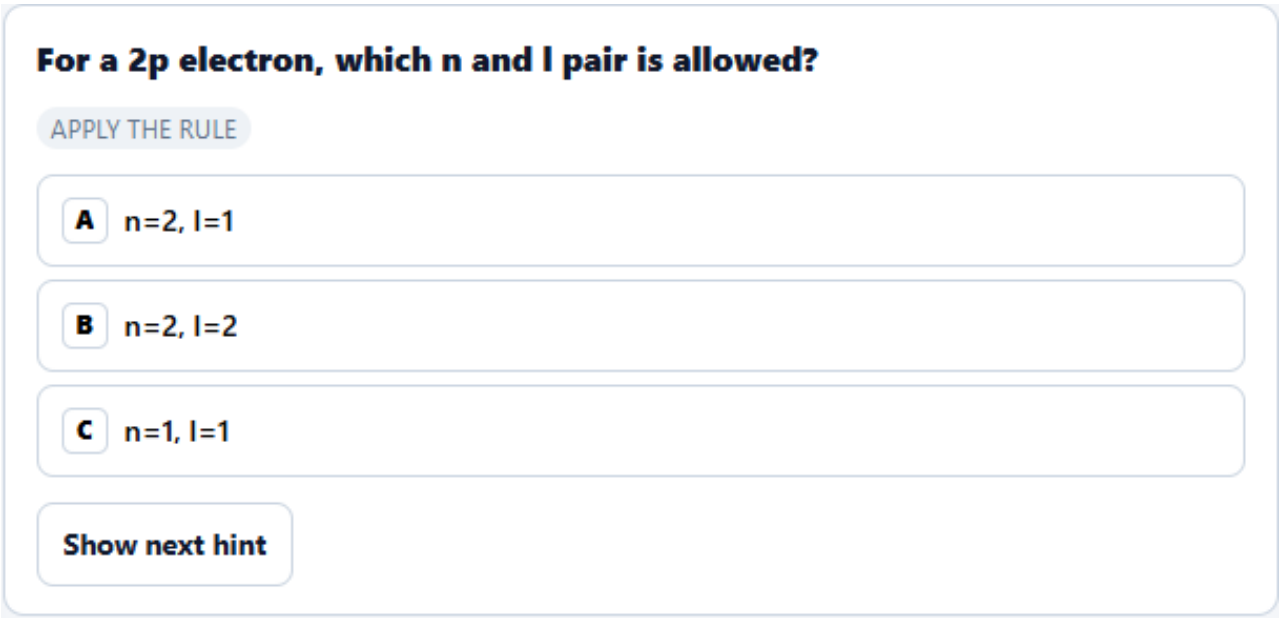
Answer type: choice

Option 0 (A): n=2, l=1

Option 1 (B): n=2, l=2

Option 2 (C): n=1, l=1

Screenshot: quantum-stage-1-question



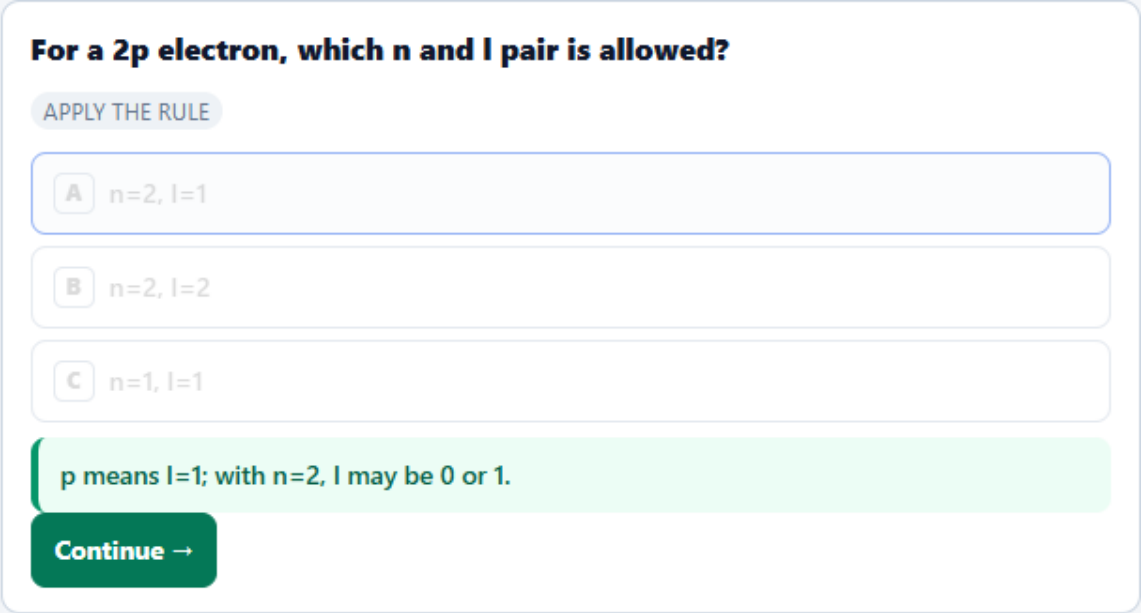

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use $l \le n-1$.

Hint 2: p is the subshell after s.

Explanation: p means l=1; with n=2, l may be 0 or 1.

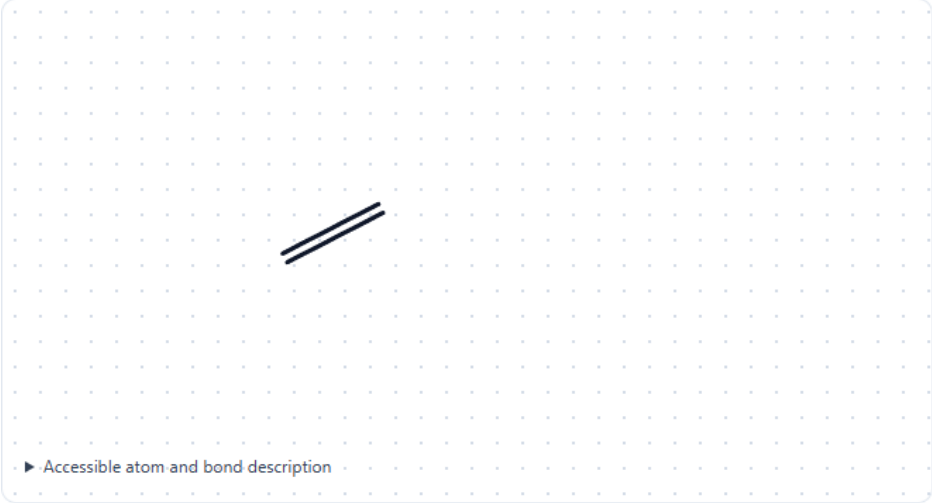
Screenshot: quantum-stage-1-feedback



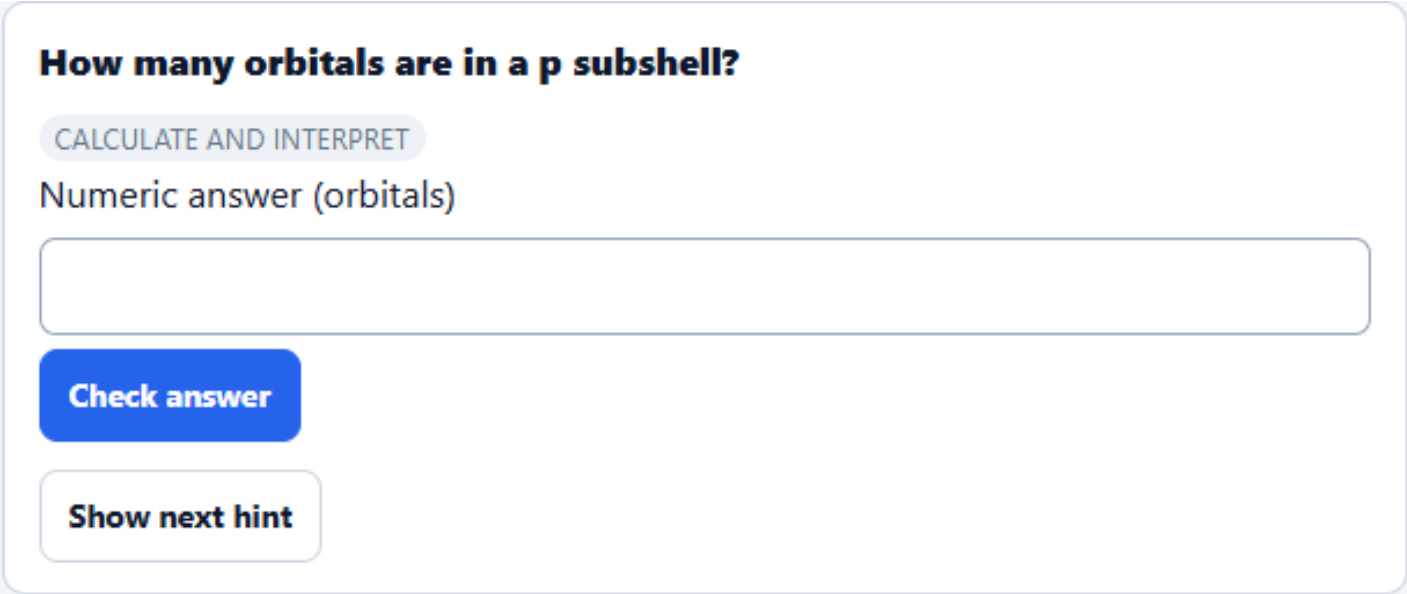

## quantum:stage-2 | How many orbitals are in a p subshell?

Answer type: number | unit: orbitals

Screenshot: quantum-stage-2-molecule



Screenshot: quantum-stage-2-question



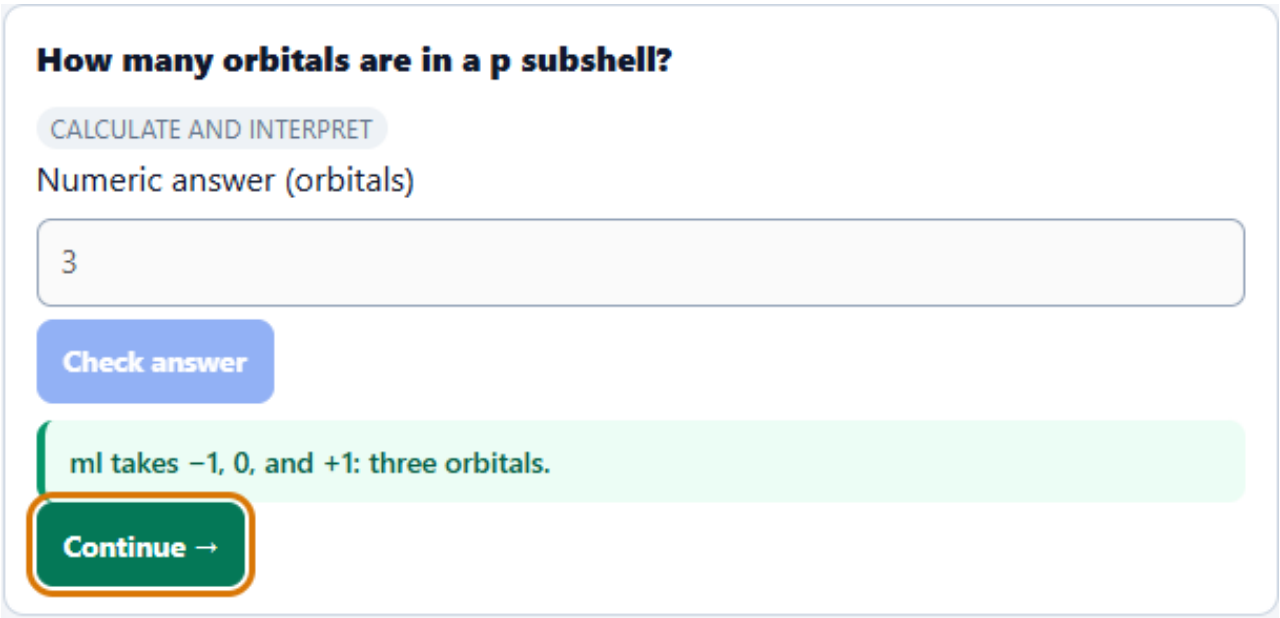

{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count the integers from −l to +l.

Hint 2: For p, l=1.

Explanation: ml takes −1, 0, and +1: three orbitals.

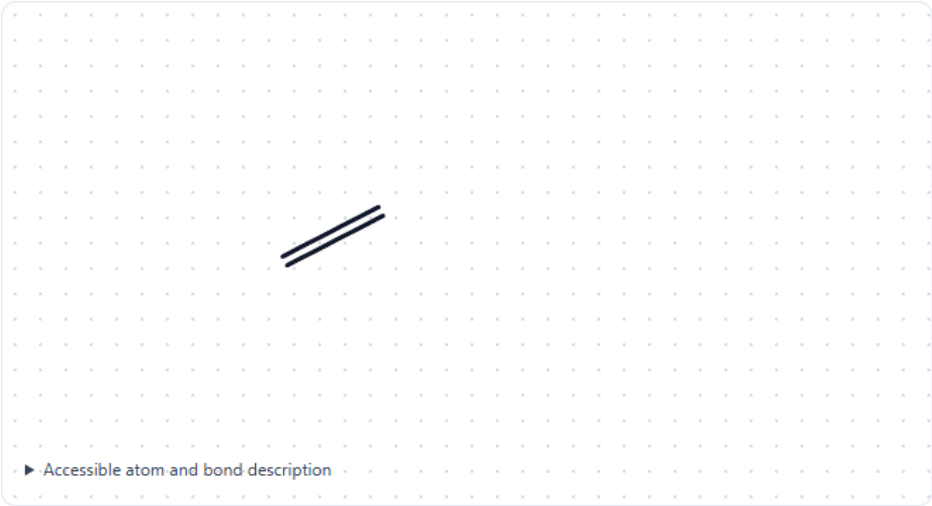
Screenshot: quantum-stage-2-feedback



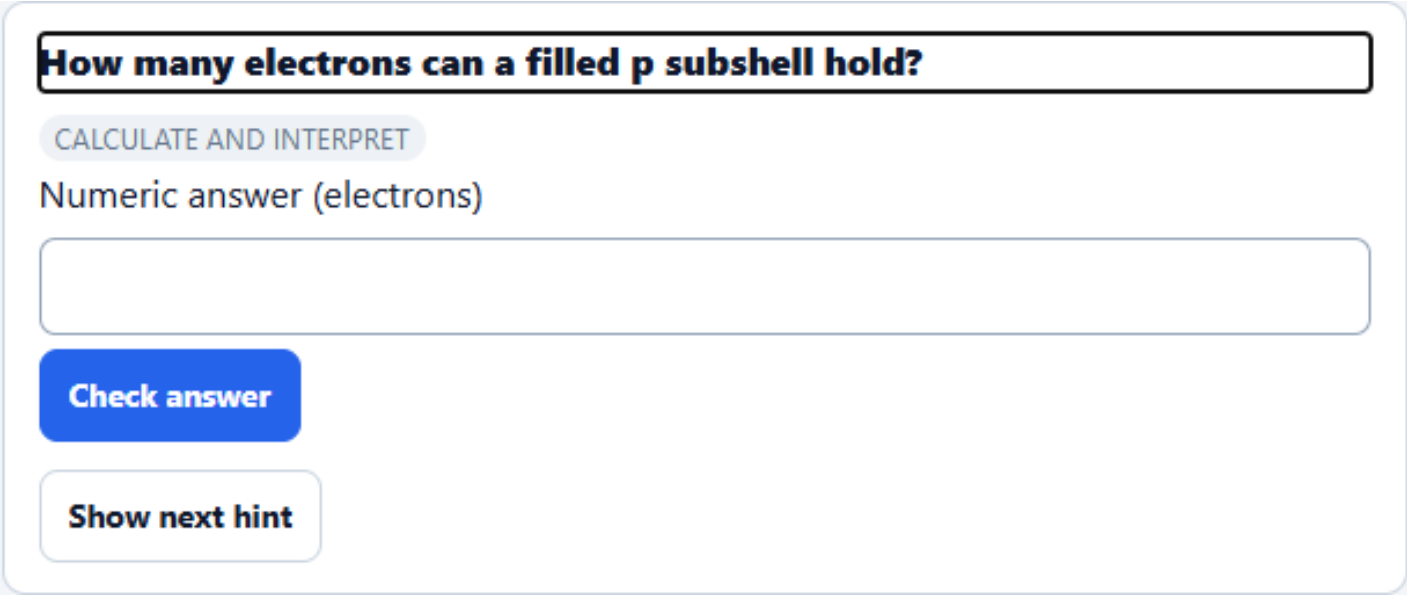

## quantum:stage-3 | How many electrons can a filled p subshell hold?

Answer type: number | unit: electrons

Screenshot: quantum-stage-3-molecule




Screenshot: quantum-stage-3-question



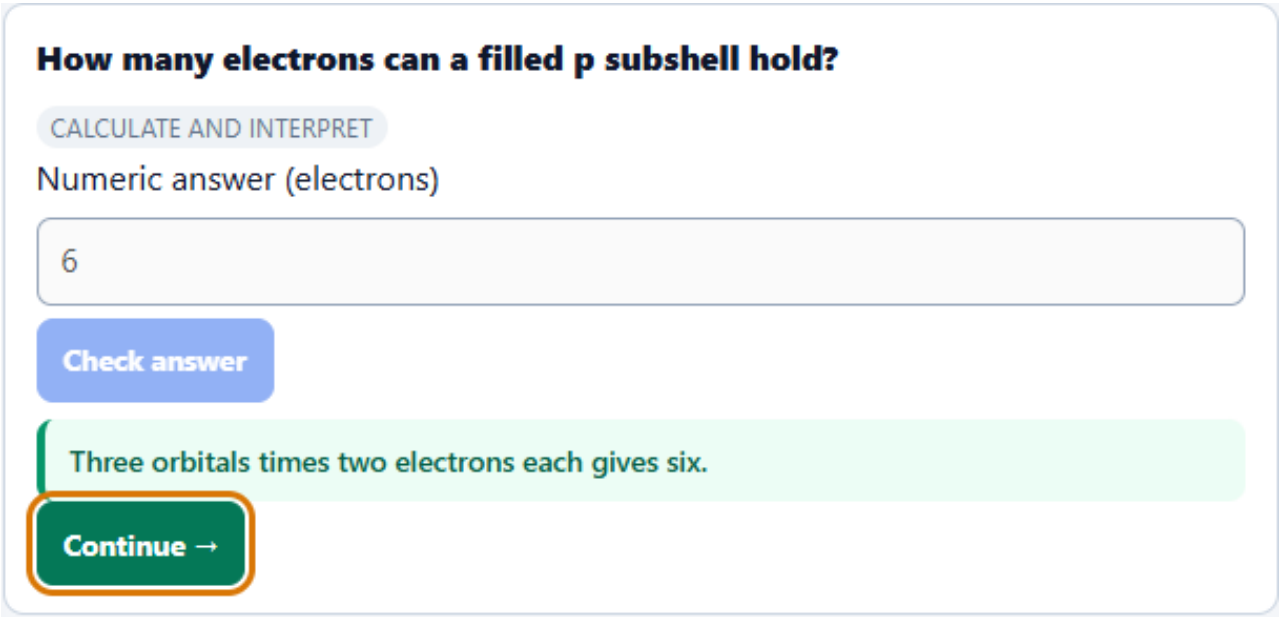

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Hint 1: Use Pauli exclusion.

Hint 2: Each orbital holds two opposite-spin electrons.

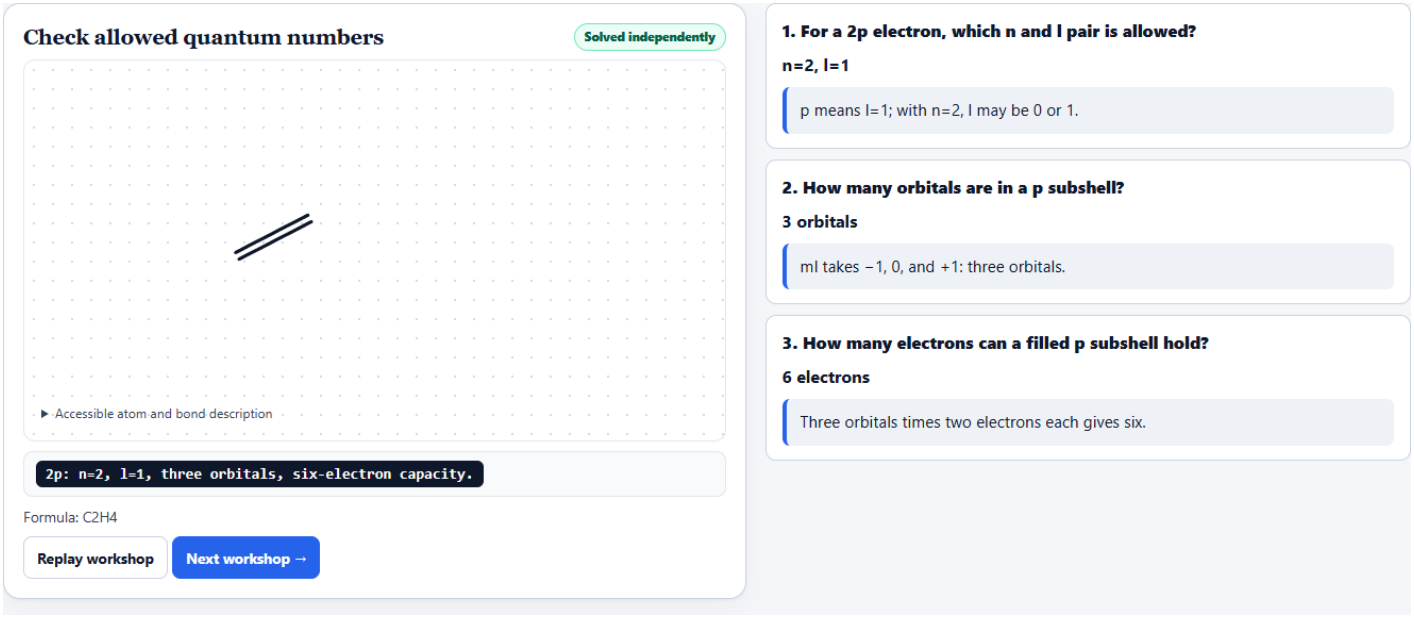
Explanation: Three orbitals times two electrons each gives six.

Screenshot: quantum-stage-3-feedback



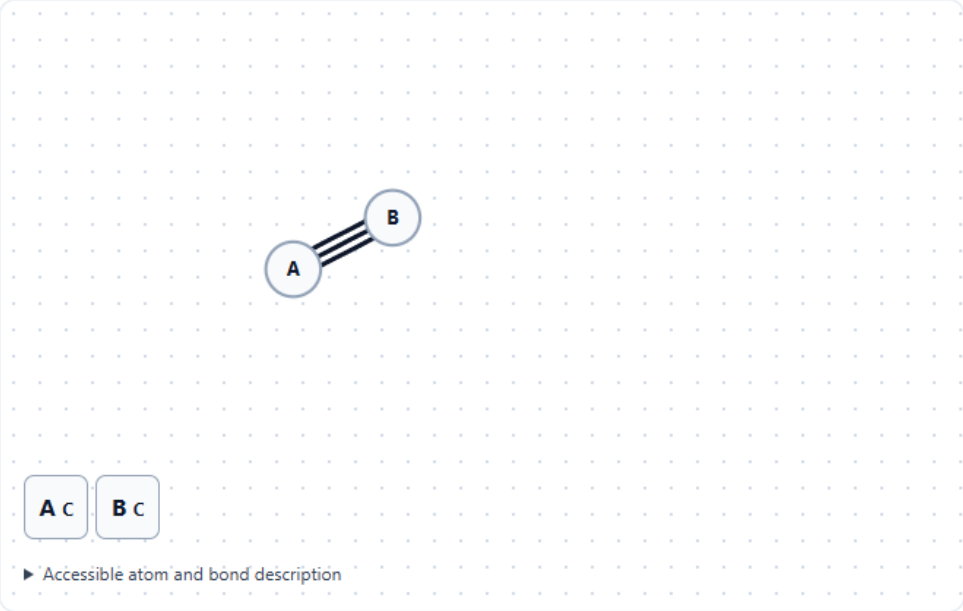

Completion summary: 2p: n=2, l=1, three orbitals, six-electron capacity.

Screenshot: quantum-complete



# bond-count | Count all sigma and pi bonds

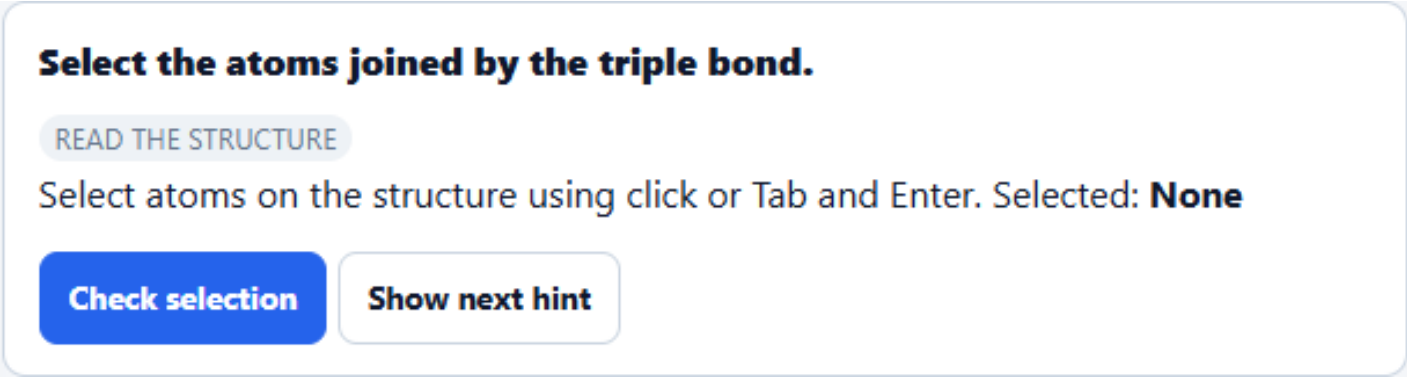
Screenshot: bond-count-stage-1-molecule




## bond-count:stage-1 | Select the atoms joined by the triple bond.

Answer type: atoms

Screenshot: bond-count-stage-1-question



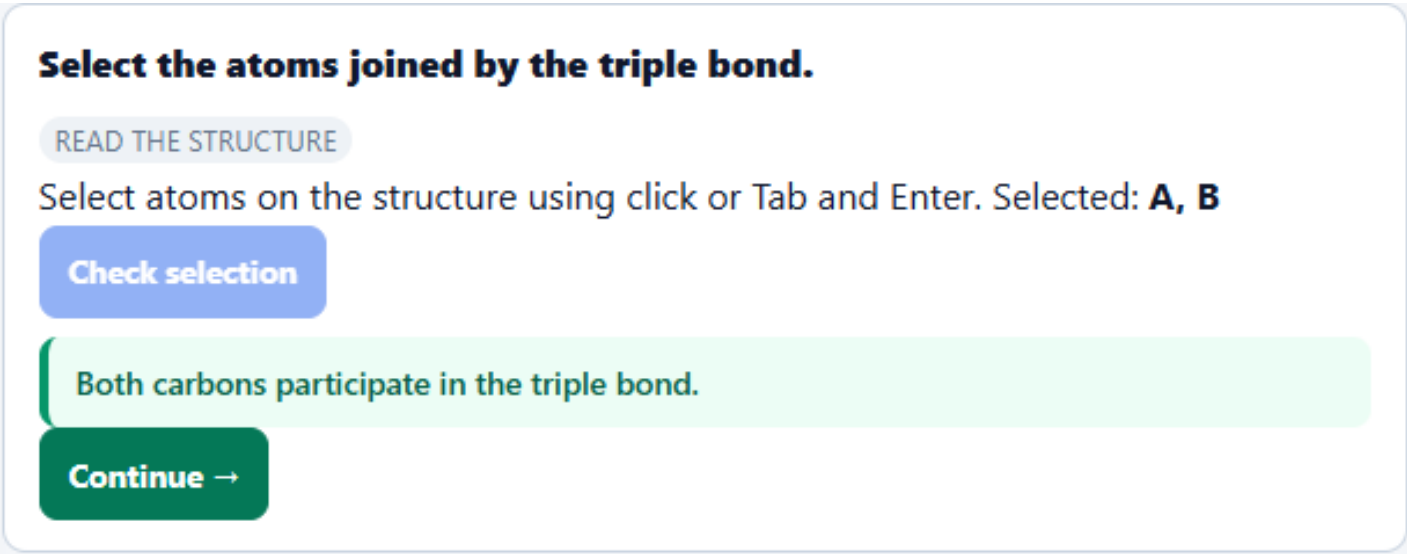

{"authorAnswer": [0, 1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 1]}}

Hint 1: Find the three parallel lines.

Hint 2: Select both ends.

Explanation: Both carbons participate in the triple bond.

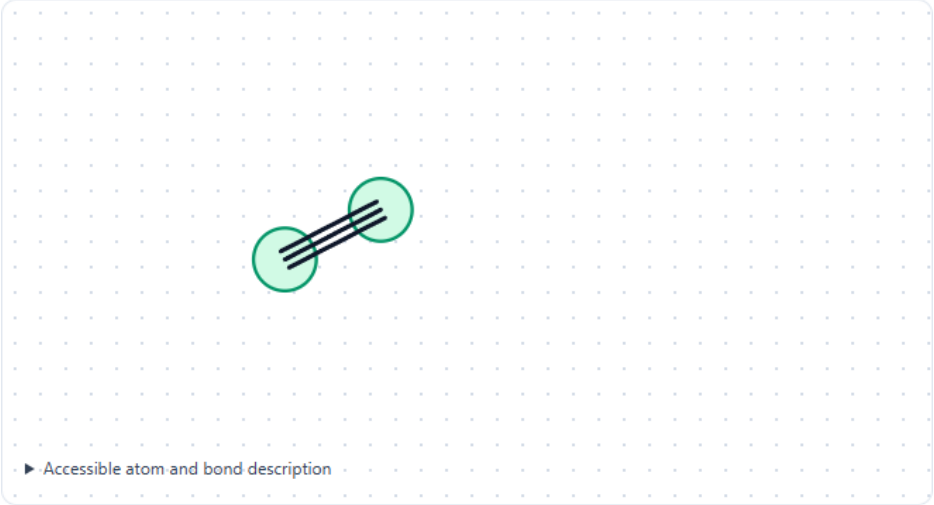
Screenshot: bond-count-stage-1-feedback



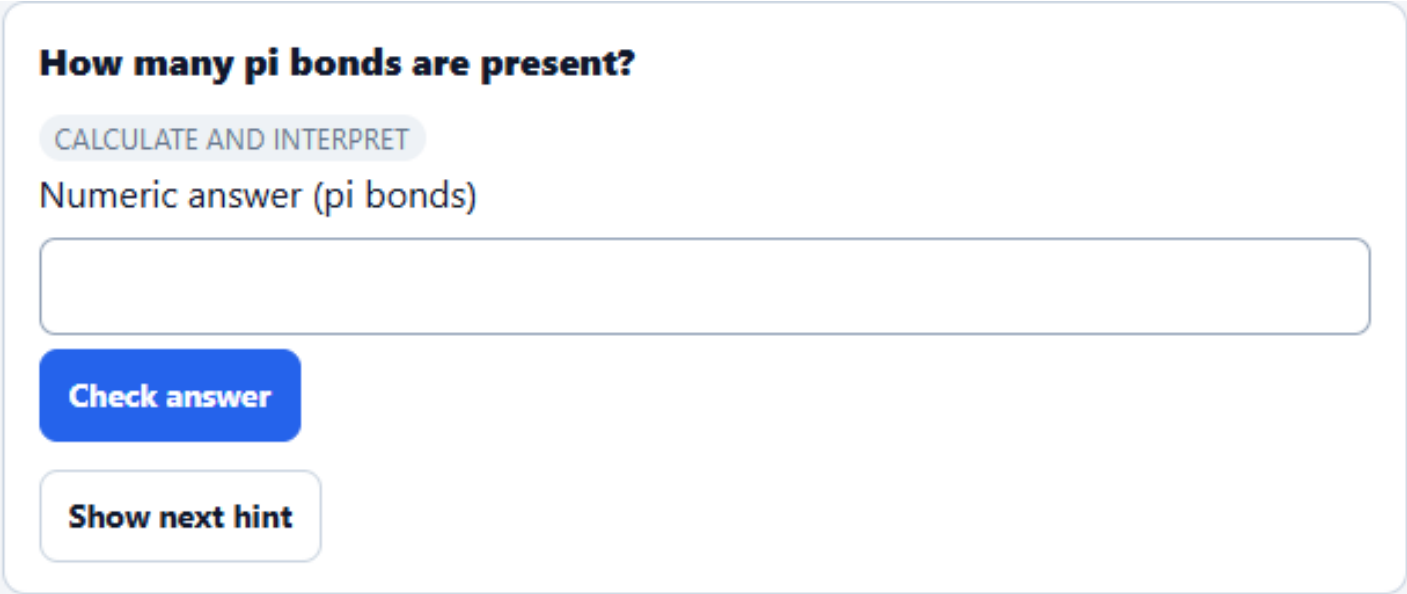

## bond-count:stage-2 | How many pi bonds are present?

Answer type: number | unit: pi bonds

Screenshot: bond-count-stage-2-molecule



Screenshot: bond-count-stage-2-question



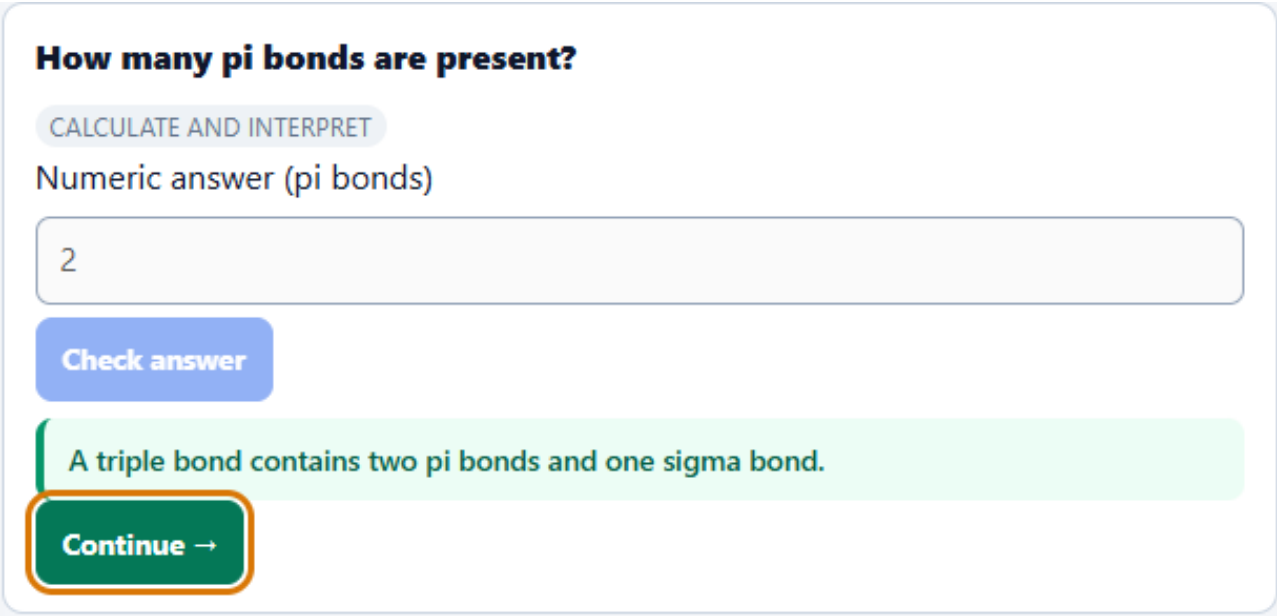

{"authorAnswer": 2, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Every bonded atom pair has one sigma bond.

Hint 2: The other two bonds are pi.

Explanation: A triple bond contains two pi bonds and one sigma bond.

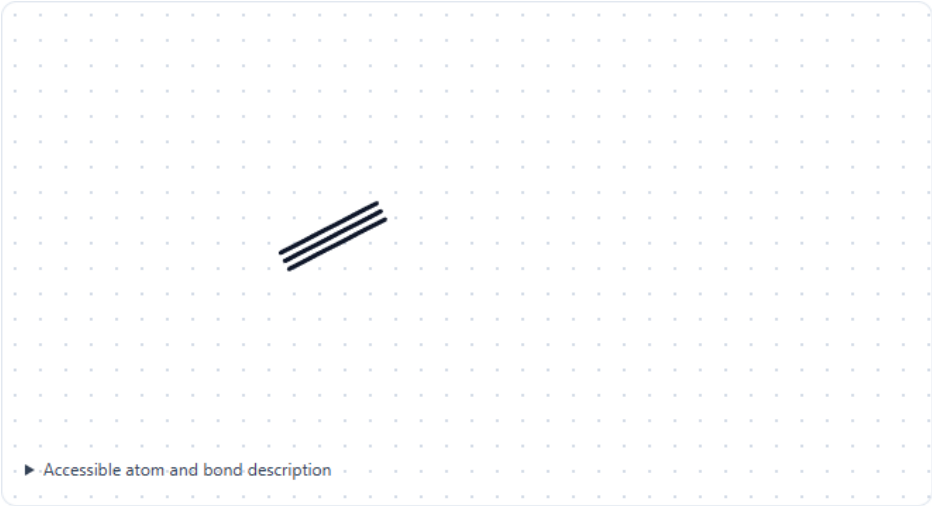
Screenshot: bond-count-stage-2-feedback



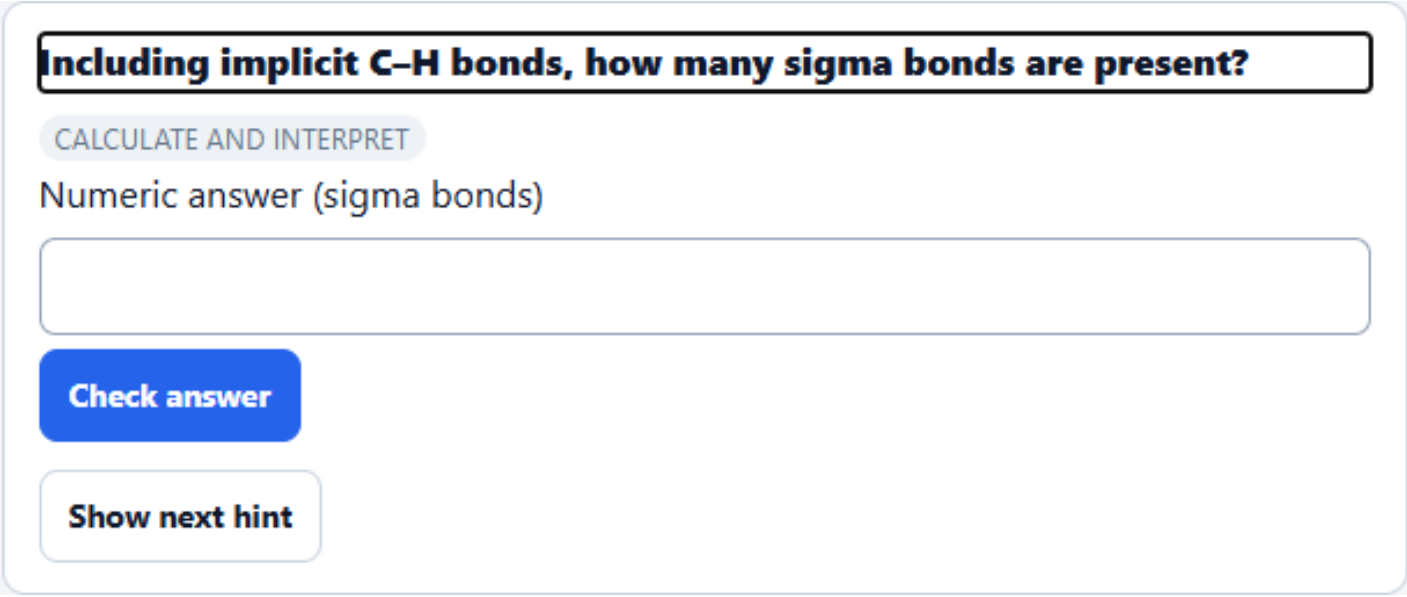

## bond-count:stage-3 | Including implicit C–H bonds, how many sigma bonds are present?

Answer type: number | unit: sigma bonds

Screenshot: bond-count-stage-3-molecule




Screenshot: bond-count-stage-3-question



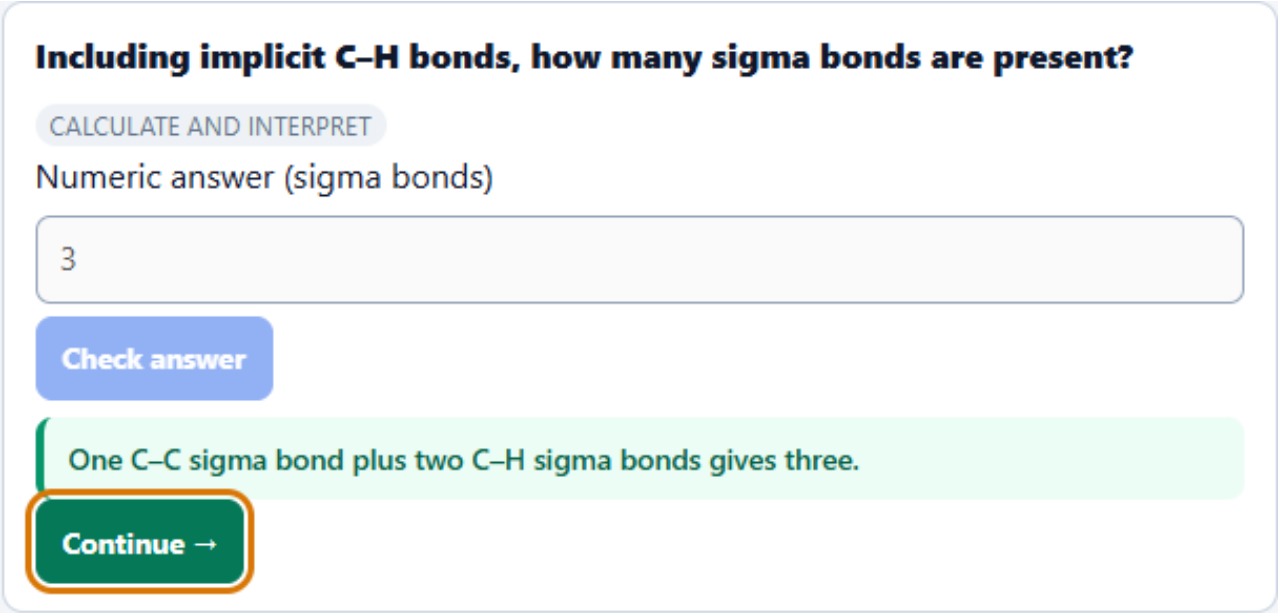

{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Each carbon has one implicit H.

Hint 2: Count the C–C sigma bond once.

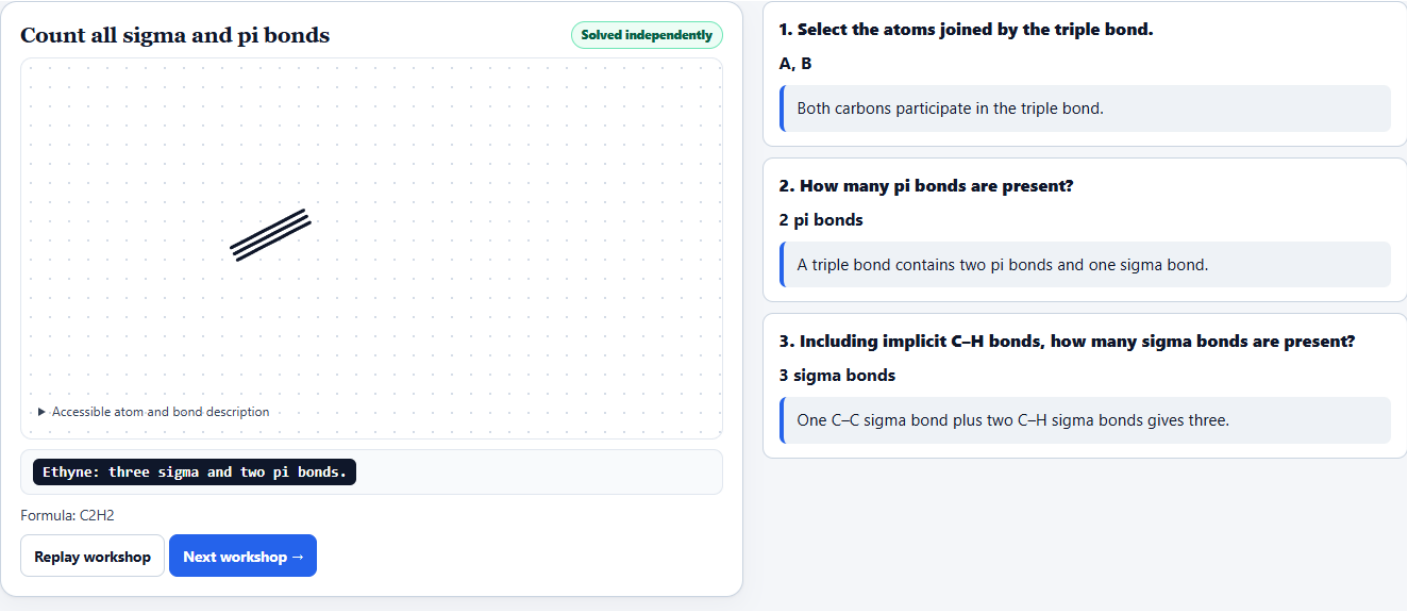
Explanation: One C–C sigma bond plus two C–H sigma bonds gives three.

Screenshot: bond-count-stage-3-feedback



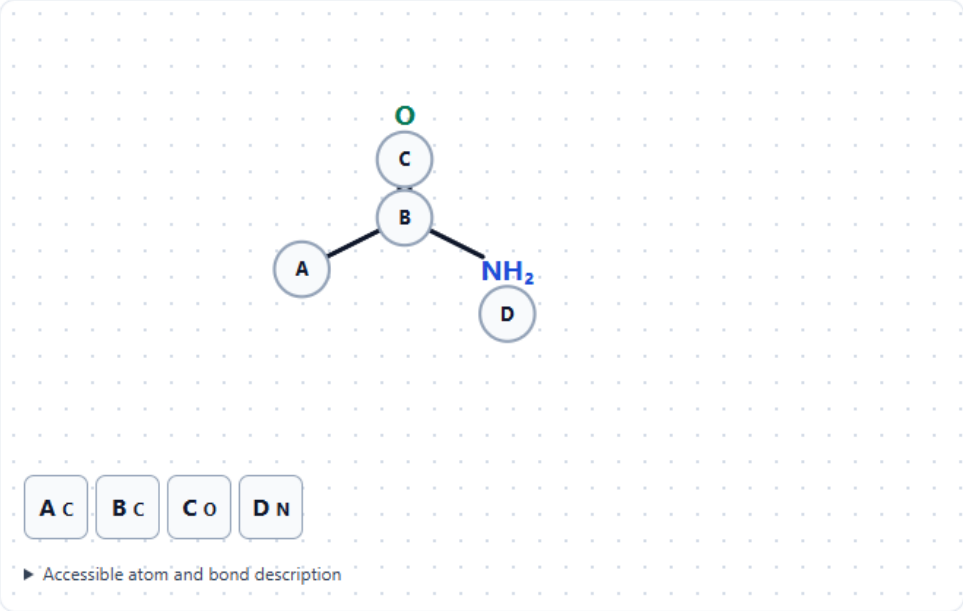

Completion summary: Ethyne: three sigma and two pi bonds.

Screenshot: bond-count-complete



# amide-resonance | Use resonance to predict geometry

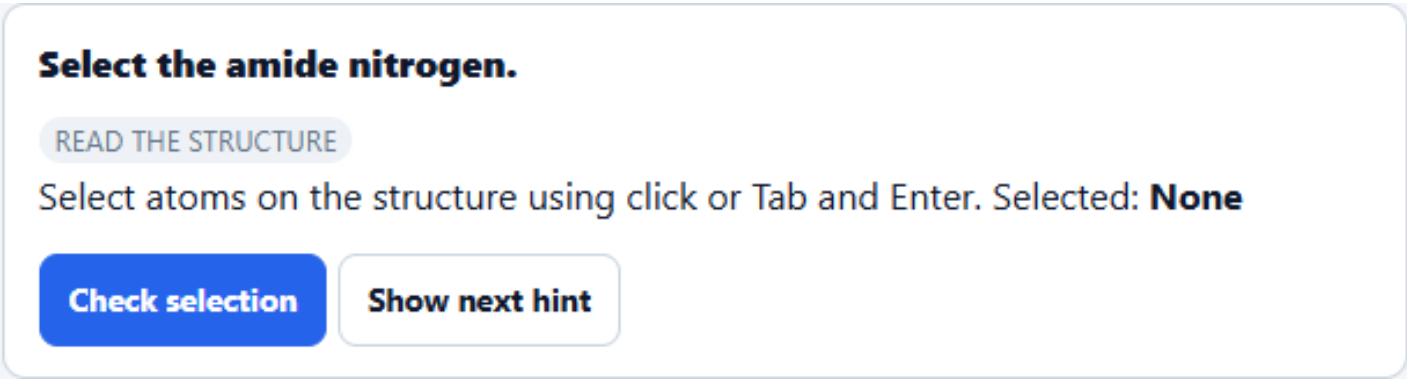
Screenshot: amide-resonance-stage-1-molecule




## amide-resonance:stage-1 | Select the amide nitrogen.

Answer type: atoms

Screenshot: amide-resonance-stage-1-question



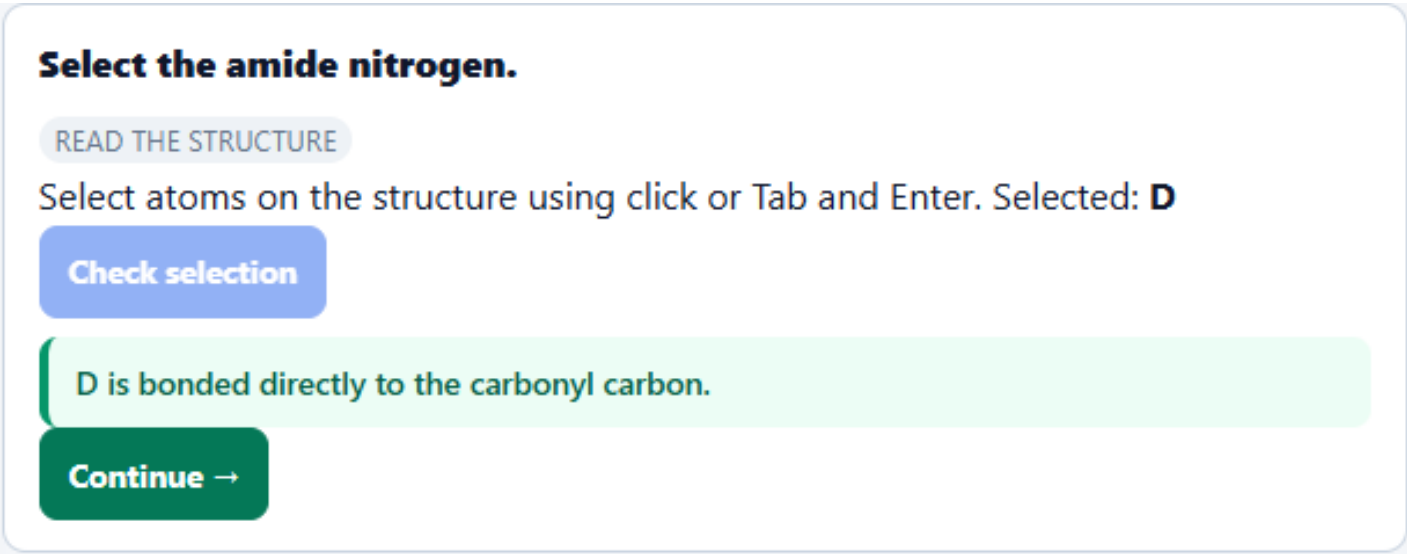

{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [3]}}

Hint 1: Read the N element label.

Hint 2: The amide N neighbors the C=O carbon.

Explanation: D is bonded directly to the carbonyl carbon.

Screenshot: amide-resonance-stage-1-feedback




## amide-resonance:stage-2 | Why is the amide group approximately planar?

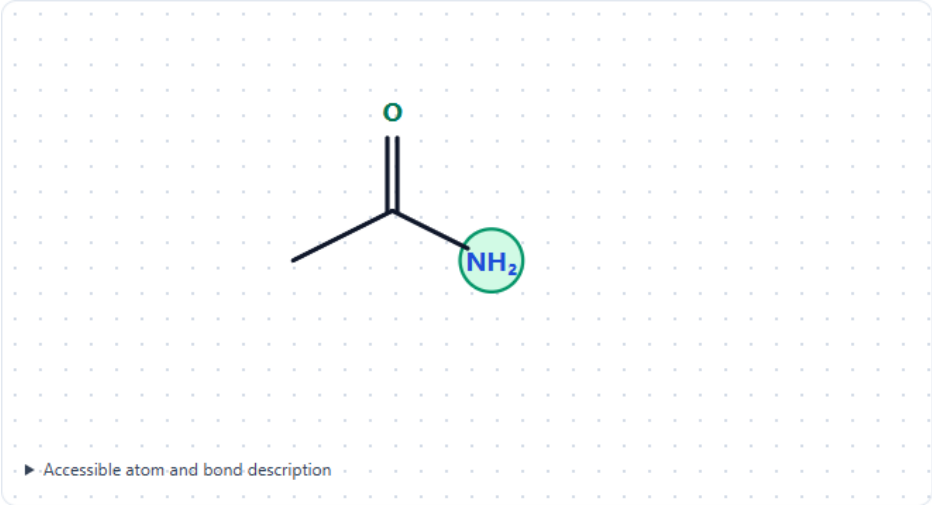
Answer type: choice

Option 0 (A): Nitrogen lone-pair conjugation with C=O

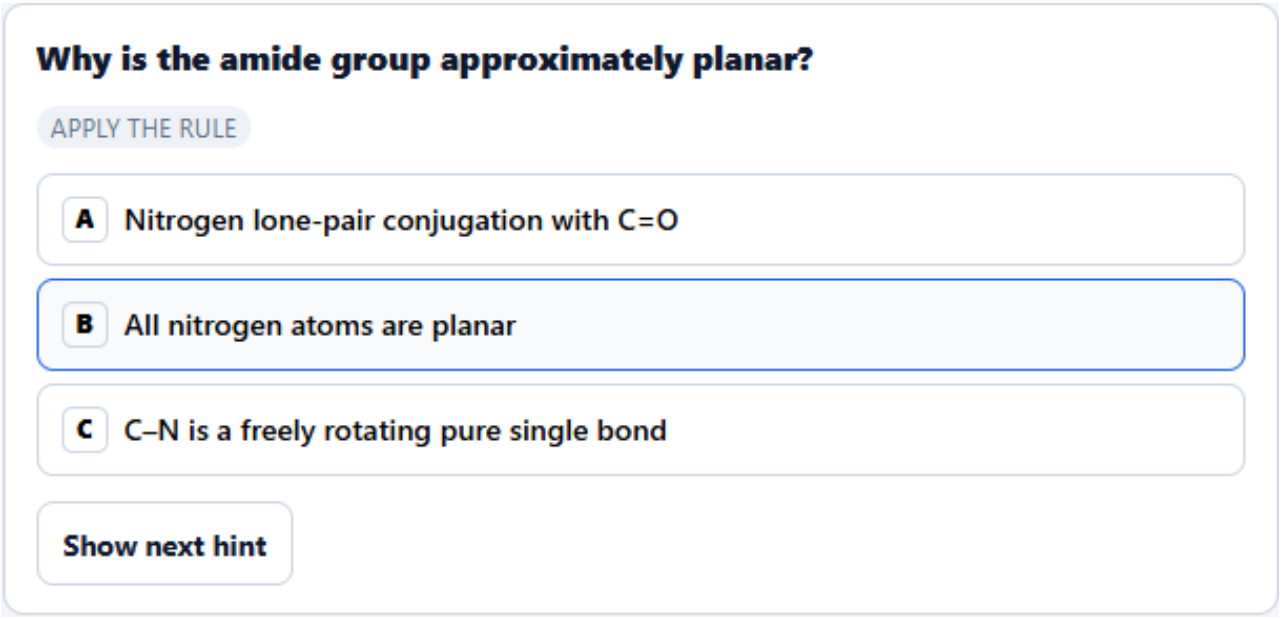
Option 1 (B): All nitrogen atoms are planar

Option 2 (C): C–N is a freely rotating pure single bond

Screenshot: amide-resonance-stage-2-molecule



Screenshot: amide-resonance-stage-2-question



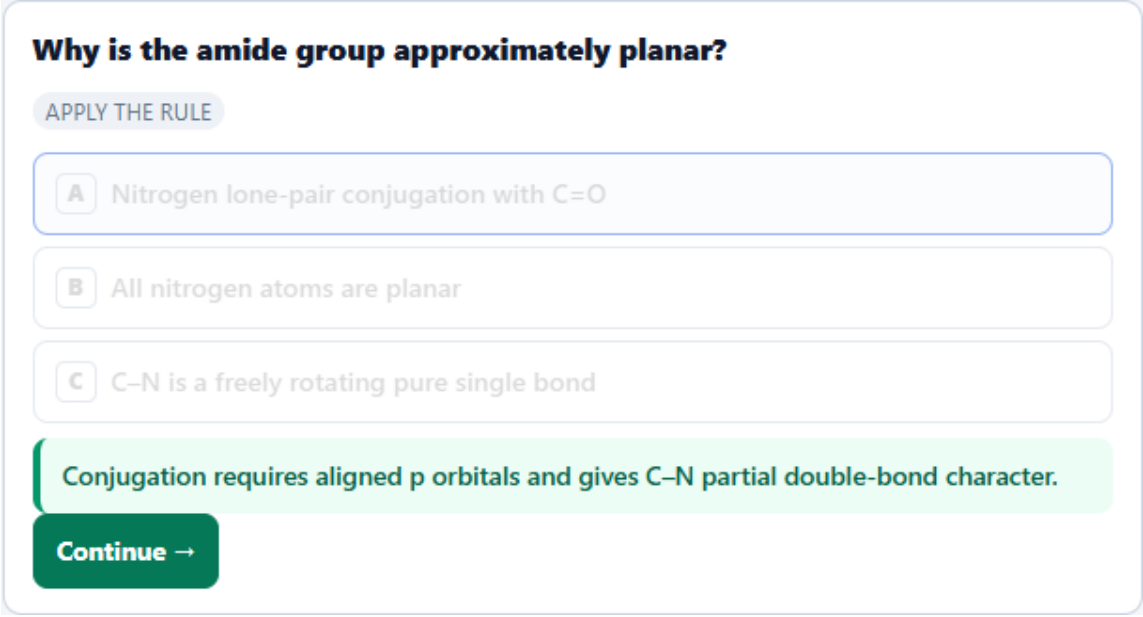

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Look for adjacent lone-pair and pi systems.

Hint 2: Delocalization requires orbital alignment.

Explanation: Conjugation requires aligned p orbitals and gives C–N partial double-bond character.

Screenshot: amide-resonance-stage-2-feedback




## amide-resonance:stage-3 | For acetate, resonance contributors differ in…

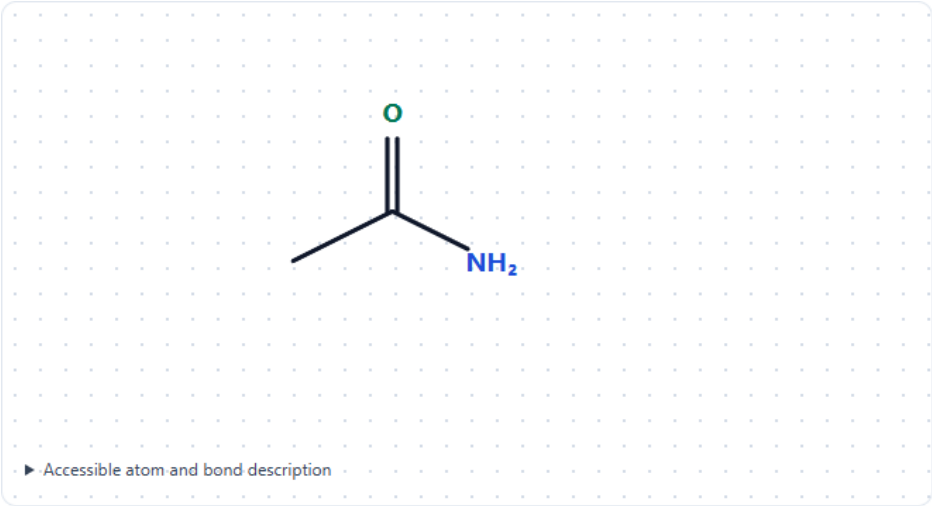
Answer type: choice

Option 0 (A): Positions of atoms

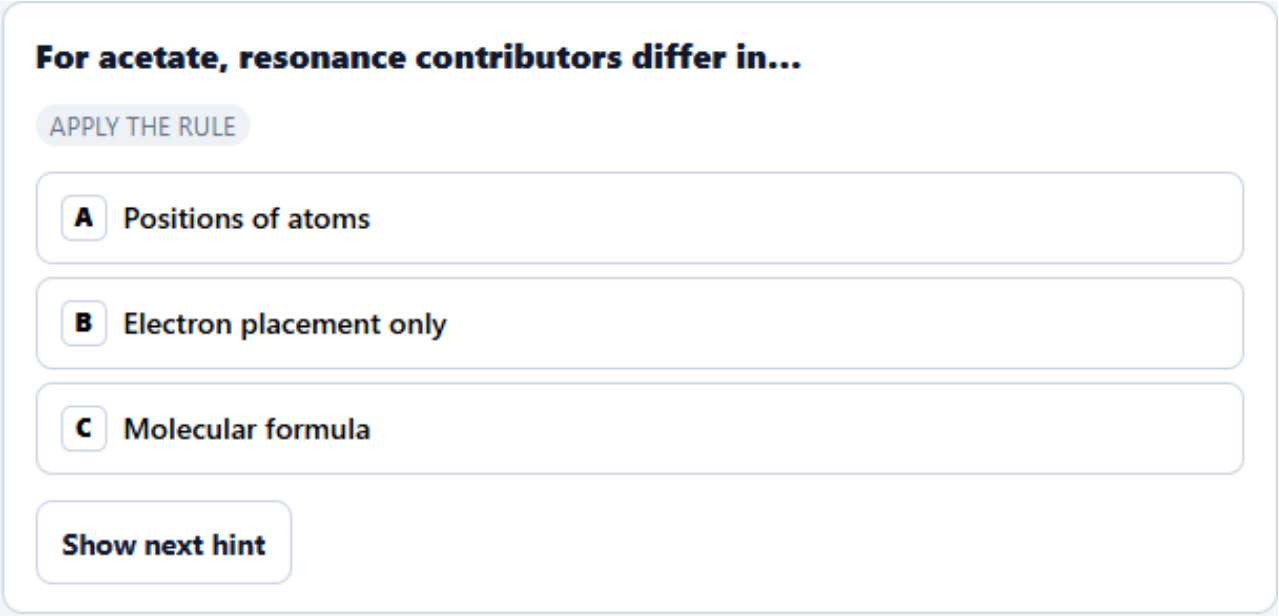
Option 1 (B): Electron placement only

Option 2 (C): Molecular formula

Screenshot: amide-resonance-stage-3-molecule




Screenshot: amide-resonance-stage-3-question



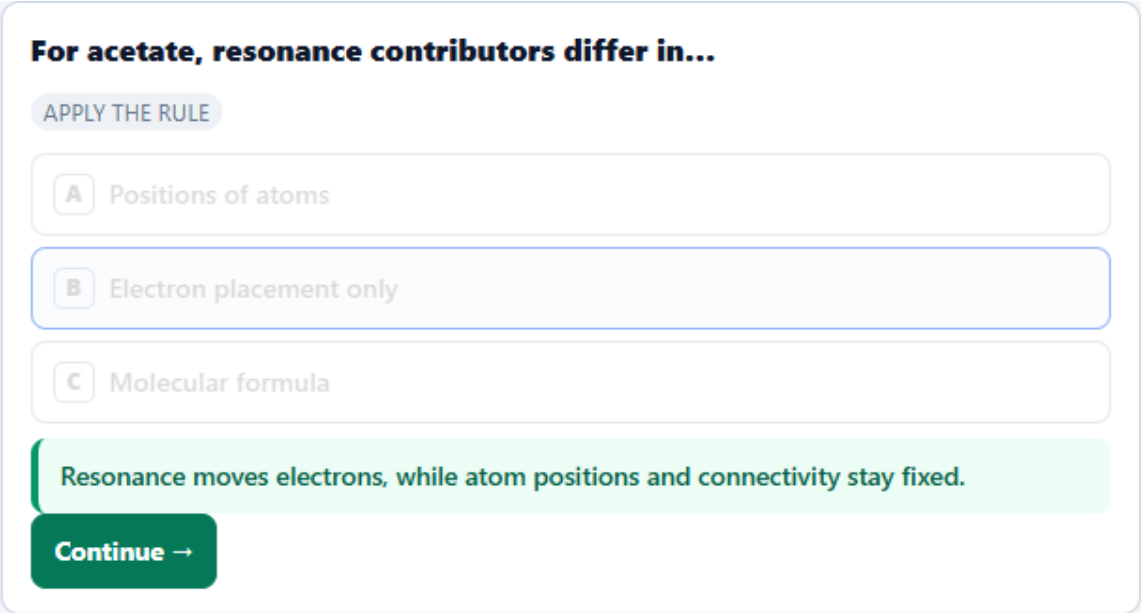

{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish resonance from isomerization.

Hint 2: The sigma framework is unchanged.

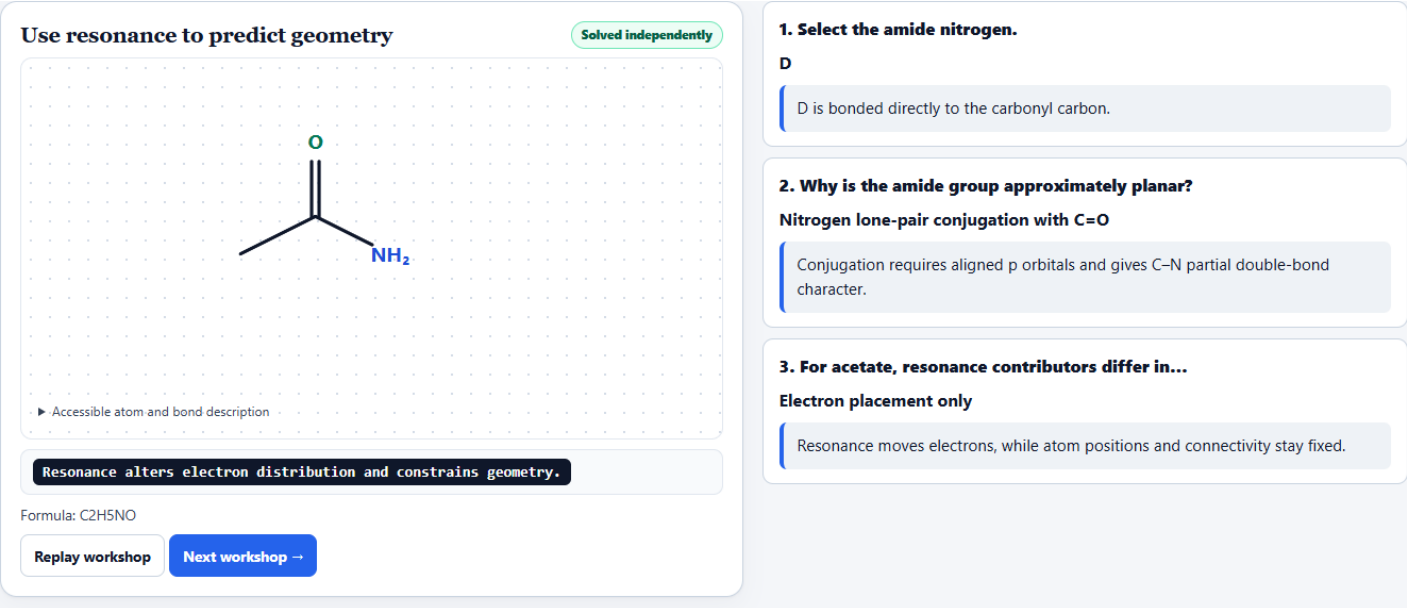
Explanation: Resonance moves electrons, while atom positions and connectivity stay fixed.

Screenshot: amide-resonance-stage-3-feedback




Completion summary: Resonance alters electron distribution and constrains geometry.

Screenshot: amide-resonance-complete



# Molecular repository: author claims to verify

## ethene

Author claims: ethene | formula C2H4

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

## ethyne

Author claims: ethyne | formula C2H2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3]]

## acetamide

Author claims: ethanamide | formula C2H5NO

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

## acetaldehyde

Author claims: ethanal | formula C2H4O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

## acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

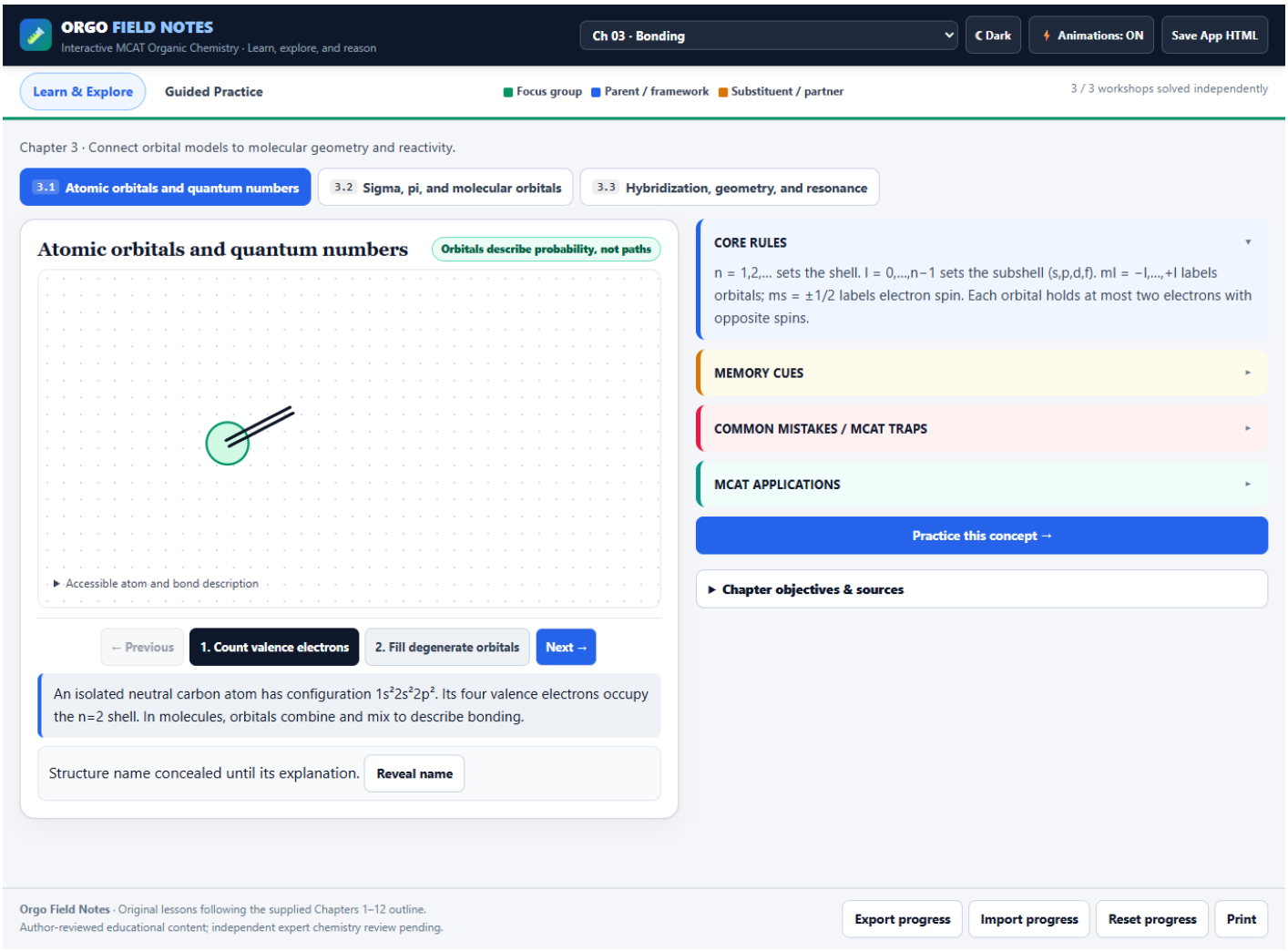
3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

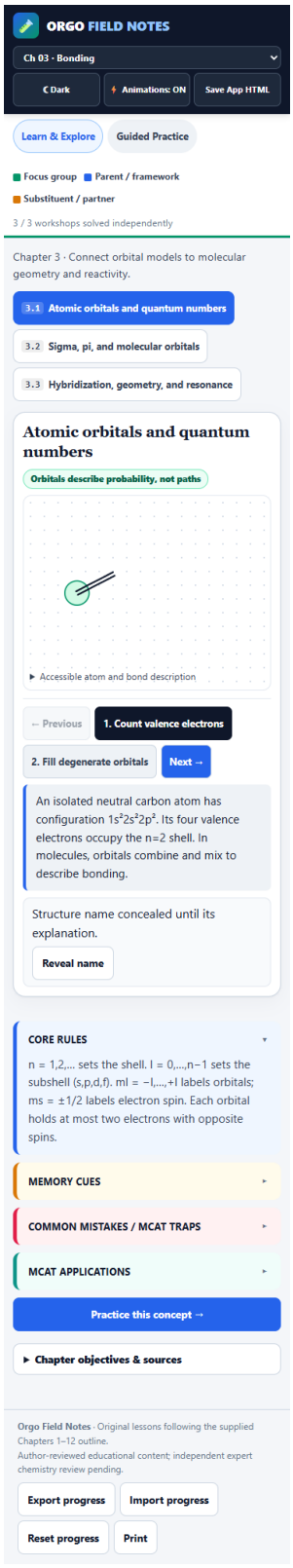
# Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

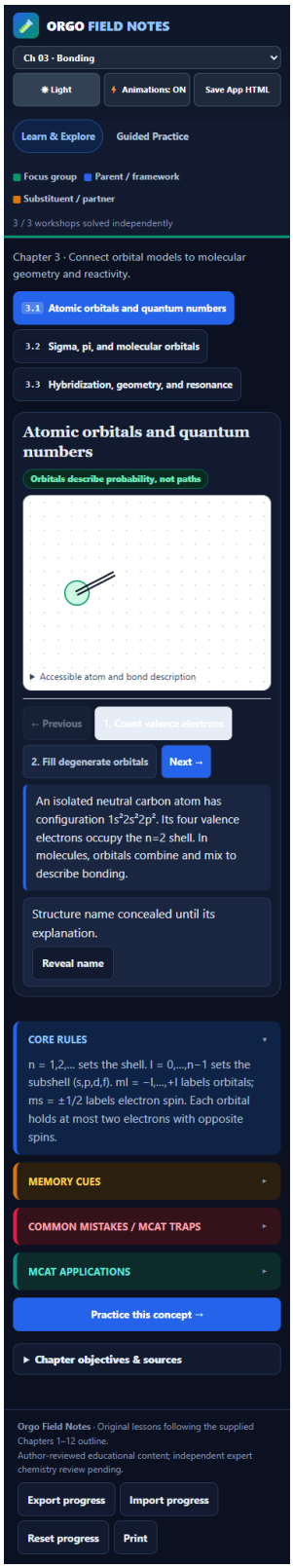
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark




# Execution evidence

41 screenshots and 10 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.